

P2 — Memory Generalization (Reviewer 452A)

CLOSED 500 accepted runs (370 A01/S01 + 130 controls) · attack A01/S01 only · retrieval (RET/L1) only

Reviewer concern

"The attack is demonstrated on an extremely simple initial memory... it's unclear whether the attack would be successful with different initial memory states... claiming that success doesn't change much from 6 to 200 records is misleading without specifying what those records are."

Conclusion (what the evidence shows)

Retrieval contamination depends on relative relevance ranking — not raw memory size — for top-k occupancy.

- Raw memory **size** did not matter for the tested **off-topic** profiles (3/60/200 records → CCR 1.0).
- **On-topic** benign content **evicts** the generic poison (CCR → 0).
- A **higher-ranking benign record evicts the poison** — confirmed causally (stronger-competitor intervention).
- Sufficiently query-aligned poison may regain top-k, but this is **template-dependent and fragile**, **NOT generally robust**.

We **revise** — not "refute" — the reviewer: composition matters; the concern is partly correct.

Run accounting (verified from raw bundles)

Campaign	Prereg	Runs	A01	Ctrl
A — Part 1 (off-topic composition variants × budgets)	PREREG_452A	200	150	50
B — composition (on-topic flood / adaptive / competitor)	PREREG_P2_memory_composition	240	170	70
C — P2b hardening (stronger competitor + 3 adaptive templates)	PREREG_P2b_hardening	60	50	10
TOTAL		500	370	130

Key results

Composition (Campaign B). Off-topic size invariant; on-topic evicts generic; adaptive (original template) survives.

cell	generic CCR	adaptive CCR	best-benign sim	
off-topic MEM003/060/200	1.0	—	0.61–0.69	CCR deterministic; controls (A00) all 0.
on-topic N=50 / 200	0.0	1.0	0.838	
on-topic N=500	0.0	0.667	0.847	

Hardening Test 1 — stronger competitor (score 0.897 > poison 0.817). Higher-ranking benign displaces then evicts.

M competitors	poison rank	top-3 CCR	poison-in-top-3 / 10 (95% CI)
0 (baseline)	1	1.0	10/10 [0.69, 1.0]
1	2	0.667	10/10 [0.69, 1.0]
3	4	0.0	0/10 [0, 0.31]

Hardening Test 2 — three frozen adaptive templates (base MEM060 + on-topic N=200); reported separately, NOT pooled.

template	rank	top-3 CCR	present / 10	verdict	
adapt_v2	1	0.667	10/10	restores (rank-1) — deterministic but fragile (+0.005)	⇒ No general-adaptive-robustness
adapt_v4	2	0.333	10/10	partial (rank-2), weak	
adapt_v3	—	0.0	0/10	FAILS to restore	

claim (1 of 3 reached rank-1; 1 failed).

Deterministic-retrieval (score audit; score = 0.6 sim + 0.2 recency + 0.2 importance)

The **poison score is fixed** (0.8171 generic / 0.8921 adaptive); recency≈1.0 and importance=0.9 are constant, so only **sim** moves the margin. Off-topic distractors score ≤ 0.76 → poison wins by +0.05–0.07; on-topic benign 0.88 > generic 0.817 → evicts; adaptive 0.892 squeaks past by +0.004–0.010. Tie-break: stable sort on 4-decimal score; equal scores keep candidate order.

Scope / limitations

A01/S01 and RET only (isolates memory composition; orthogonal to the threat-model work). Off-topic "size" = 3 discrete profiles (no MEM1000); one on-topic template; local deterministic embedder. CCR = observed values; binary "poison present in top-k" = exact Clopper-Pearson 95% CIs.